



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer Science
Faculty of Science and Technology**

CSC2210: OBJECT ORIENTED PROGRAMMING 2

SPRING 2024-2025

Section: [X]

Group No: 05

Project Report On

Project Name [Safe Home Authority]

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CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				Total =
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

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Introduction:

Safe Home Authority is a desktop-based application developed using **C#** in **Visual Studio Code**, aimed at improving residential safety and managing tenant information in a structured and secure way.

The system introduces two primary user roles: **Landlords** and **Police Officers**. Landlords are authorized to input and manage information only for the tenants residing in their own buildings. This restriction ensures that no landlord can access or modify data outside their property, maintaining data privacy and ownership integrity.

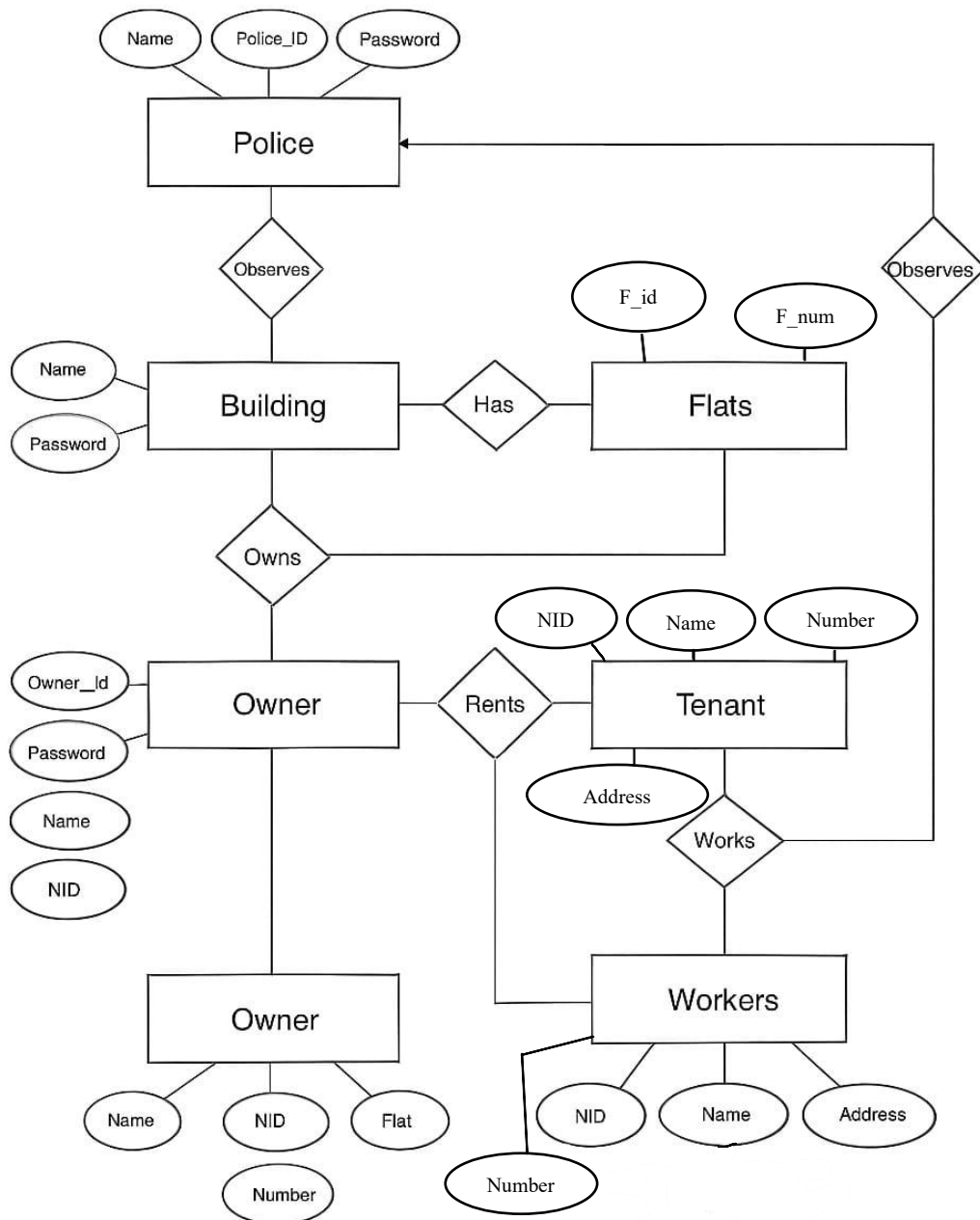
On the other hand, Police Officers have access to a dedicated admin login panel. Once logged in, they can instantly view all tenant and flat information across registered buildings—without needing to perform individual NID-based searches. This feature provides law enforcement with quick and reliable access to verified resident data, which can be crucial for maintaining community security and supporting investigations.

By clearly separating access roles and streamlining data visibility for police, the **Safe Home Authority** system not only enhances accountability and safety but also builds a digital bridge between landlords and law enforcement agencies.

User Story:

Police officers and building owners must log in to the Safe Home Authority system using secure credentials to access their respective dashboards. Building owners have access only to their own building accounts, where they can input, update, and manage detailed information about their tenants. Police officers, on the other hand, log in through an admin panel that grants them the authority to search and view tenant information across all registered buildings and flats by using flat IDs, building IDs, or National ID numbers. This dual-login system ensures data security and privacy while enabling effective communication and cooperation between landlords and law enforcement. Ultimately, the system helps maintain accurate tenant records and supports the police in verifying identities and enhancing community safety during investigations or routine checks.

ER Diagram:



Normalization:

UNF (Unnormalized Form)

Owner: o_id, o_name, o_pass, o_mobile, Buildings: [b_id, b_name, city, sector], ...

Tenant: t_id, t_name, f_id, [city, sector]

Worker: w_id, w_name, mobile, [city, sector], Flats: [f_id, f_name]

1NF (First Normal Form)

Repeating groups removed

Each field is atomic

Separate rows for each building, tenant, worker assignment

Tables: owner, building, flat, worker, tenant, works

2NF (Second Normal Form)

Partial dependencies removed

Composite keys (e.g., works) refer only to related foreign keys

All non-key attributes fully depend on whole primary key

f_name moved to flat table

3NF (Third Normal Form) – Finalized Form

city and sector removed from tenant and worker (inferred via f_id → flat → building)

Final normalized tables:

- police
- owner
- own
- building
- flat
- tenant
- worker
- works
- address

Tables

1. Police Table

```
CREATE TABLE [dbo].[police](
    [p_id] [nchar](10) NOT NULL,
    [p_name] [nchar](10) NOT NULL,
    [jobtitle] [nvarchar](max) NOT NULL,
    CONSTRAINT [PK_police] PRIMARY KEY CLUSTERED
    (
        [p_id] ASC
    )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON) ON [PRIMARY]
) ON [PRIMARY] TEXTIMAGE_ON [PRIMARY]
```

	COLUMN_NAME	DATA_TY...	CHARACTER_MAXIMUM LENG...	IS_NULLAB...	COLUMN_DEFAU...
1	p_id	nchar	10	NO	NULL
2	p_name	nchar	10	NO	NULL
3	jobtitle	nvarchar	-1	NO	NULL

2. Owner Table

```
CREATE TABLE [dbo].[owner](
    [o_id] [nchar](10) NOT NULL,
    [o_name] [nvarchar](50) NOT NULL,
    [o_pass] [nchar](10) NOT NULL,
    [o_mobile] [numeric](18, 0) NOT NULL,
    CONSTRAINT [PK_owner] PRIMARY KEY CLUSTERED
(
    [o_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS
= ON, ALLOW_PAGE_LOCKS = ON) ON [PRIMARY]
) ON [PRIMARY]
```

	COLUMN_NAME	DATA_TY...	CHARACTER_MAXIMUM LENG...	IS_NULLAB...	COLUMN_DEFAU...
1	o_id	nchar	10	NO	NULL
2	o_name	nvarchar	50	NO	NULL
3	o_pass	nchar	10	NO	NULL
4	o_mobile	numeric	NULL	NO	NULL

3. Own Table

```
CREATE TABLE [dbo].[own](
    [b_id] [nchar](10) NOT NULL,
    [b_name] [nvarchar](max) NULL,
    [city] [nvarchar](max) NULL,
    [sector] [nvarchar](max) NULL,
    [o_id] [nchar](10) NULL,
    CONSTRAINT [PK_own] PRIMARY KEY CLUSTERED
(
    [b_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY
= OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON) ON [PRIMARY]
) ON [PRIMARY] TEXTIMAGE_ON [PRIMARY]
```

	COLUMN_NAME	DATA_TY...	CHARACTER_MAXIMUM LENG...	IS_NULLAB...	COLUMN_DEFAU...
1	b_id	nchar	10	NO	NULL
2	b_name	nvarchar	-1	YES	NULL
3	city	nvarchar	-1	YES	NULL
4	sector	nvarchar	-1	YES	NULL
5	o_id	nchar	10	YES	NULL

4. Tenant Table

```

CREATE TABLE [dbo].[tenant](
    [t_id] [nchar](10) NOT NULL,
    [t_name] [nchar](10) NOT NULL,
    [city] [nchar](10) NOT NULL,
    [sector] [nchar](10) NULL,
    [f_id] [nchar](10) NOT NULL,
    CONSTRAINT [PK_tenant] PRIMARY KEY CLUSTERED
(
    [t_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY
= OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON) ON [PRIMARY]
) ON [PRIMARY]

```

	COLUMN_NAME	DATA_TY...	CHARACTER_MAXIMUM LENG...	IS_NULLAB...	COLUMN_DEFAU...
1	t_id	nchar	10	NO	NULL
2	t_name	nchar	10	NO	NULL
3	city	nchar	10	NO	NULL
4	sector	nchar	10	YES	NULL
5	f_id	nchar	10	NO	NULL

5. Worker Table

```

CREATE TABLE [dbo].[workers](
    [w_id] [nchar](10) NOT NULL,
    [w_name] [nchar](10) NOT NULL,
    [w_number] [numeric](18, 0) NOT NULL,
    [city] [nchar](10) NOT NULL,
    [sector] [nchar](10) NOT NULL,
    [f_id] [nchar](10) NOT NULL,
    CONSTRAINT [PK_workers] PRIMARY KEY CLUSTERED
(
    [w_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY
= OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON) ON [PRIMARY]
) ON [PRIMARY]

```

	COLUMN_NAME	DATA_TY...	CHARACTER_MAXIMUM LENG...	IS_NULLAB...	COLUMN_DEFAU...
1	w_id	nchar	10	NO	NULL
2	w_name	nchar	10	NO	NULL
3	w_number	numeric	NULL	NO	NULL
4	city	nchar	10	NO	NULL
5	sector	nchar	10	NO	NULL
6	f_id	nchar	10	NO	NULL

6. Flat Table

```

CREATE TABLE [dbo].[flat](
    [f_id] [nchar](10) NOT NULL,
    [f_number] [nchar](10) NOT NULL,
    [b_id] [nchar](10) NOT NULL,
    CONSTRAINT [PK_flat] PRIMARY KEY CLUSTERED
    (
        [f_id] ASC
    )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY
    = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON) ON [PRIMARY]
) ON [PRIMARY]

```

	COLUMN_NAME	DATA_TY...	CHARACTER_MAXIMUM LENG...	IS_NULLAB...	COLUMN_DEFAU...
1	f_id	nchar	10	NO	NULL
2	f_number	nchar	10	NO	NULL
3	b_id	nchar	10	NO	NULL

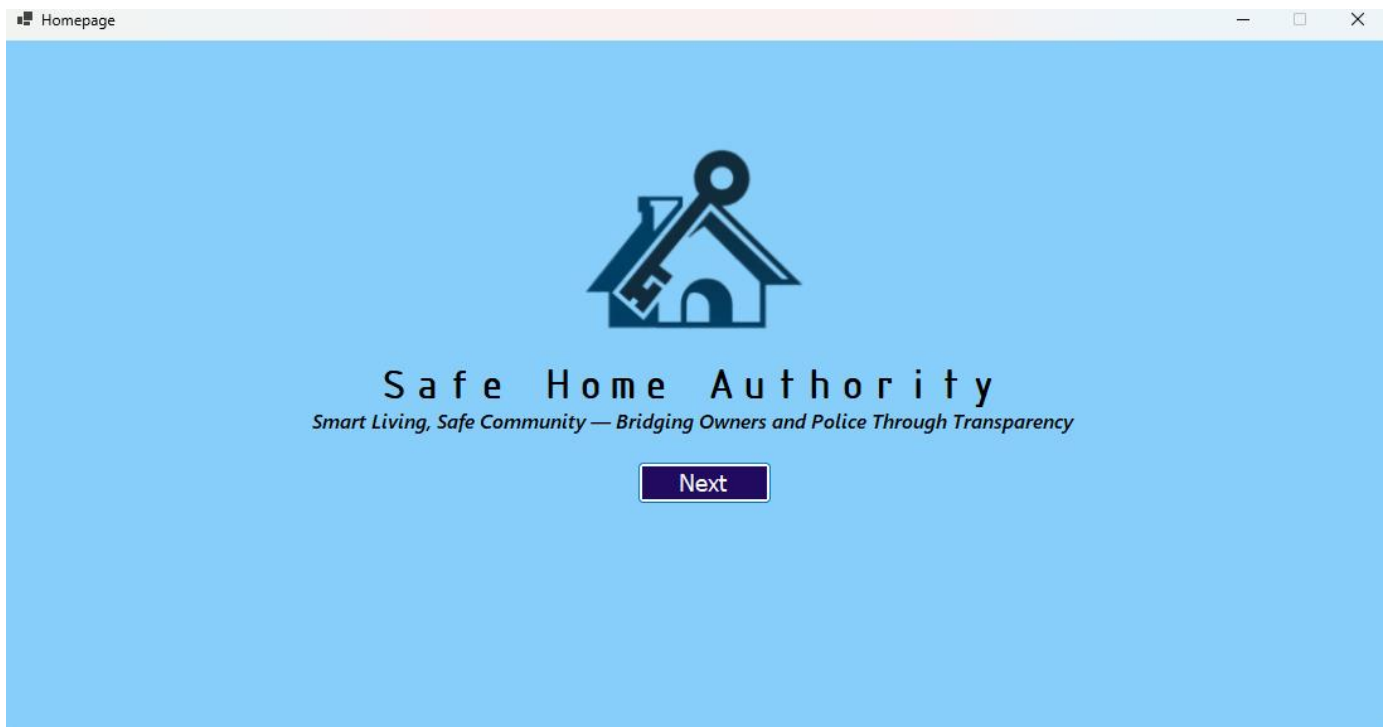
7. Building Table

```

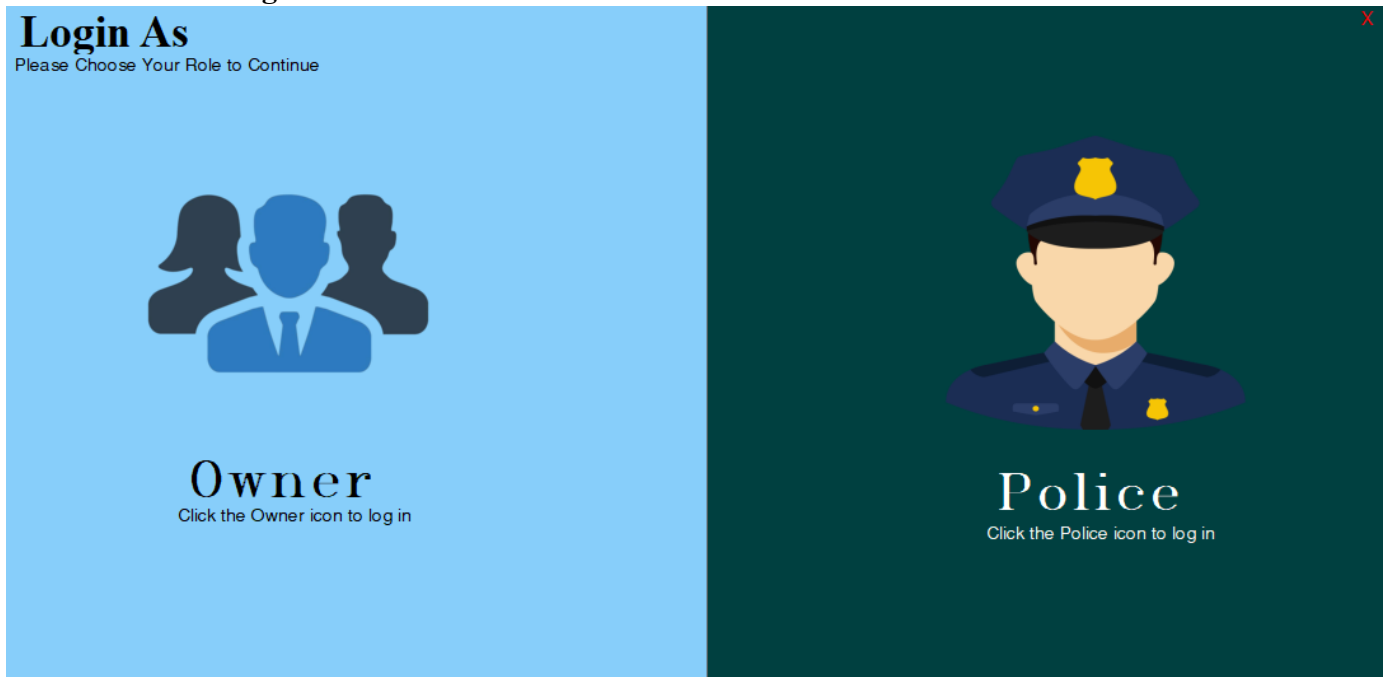
CREATE TABLE [dbo].[building](
    [b_id] [nchar](10) NOT NULL,
    [b_name] [nchar](10) NOT NULL,
    [city] [nvarchar](max) NULL,
    [sector] [nchar](10) NULL,
    CONSTRAINT [PK_building_1] PRIMARY KEY CLUSTERED
    (
        [b_id] ASC
    )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY
    = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON) ON [PRIMARY]
) ON [PRIMARY] TEXTIMAGE_ON [PRIMARY]

```

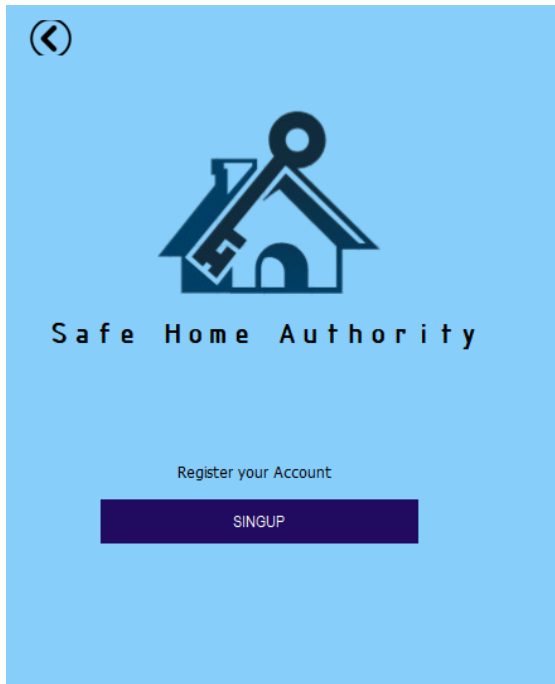
	COLUMN_NAME	DATA_TY...	CHARACTER_MAXIMUM LENG...	IS_NULLAB...	COLUMN_DEFAU...
1	b_id	nchar	10	NO	NULL
2	b_name	nchar	10	NO	NULL
3	city	nvarchar	-1	YES	NULL
4	sector	nchar	10	YES	NULL



8. Login AS Owner or Police Form



9. Owner Login Form



X

Owner Login Account

User Id:

Password:

☐ Show Password

LOGIN

10. Owner Registration Form



X

Register Account

User Id:

Password:

☐ Show Password

User Name:

Mobile No:

SINGUP

11. Owner Dashboard



Safe Home Authority

- RegisterBuilding
- Show Information
- Add Tenant
- Add Worker
- Developers

OWNER DASHBOARD

Manage Your Building

Building Id:

Manage

List Of Your Buildings

	Owner ID	Name	Building ID	Building Name	Building Address City	Building Address Sector
▶	2020	Tamim	07	AIUB	Dhaka	02
*						

12. Register Building



Safe Home Authority

BUILDING MANAGEMENT

Building Id:

Building Name:

City:

Sector:

Register

13. Flat Management



FLAT MANAGEMENT

FLAT ID

FLAT NAME

ADD

UPDATE

DELETE

List Of Your Flats

	f_id	f_number	b_id
▶	1A	East	07
	1B	North	07
*			

14. Add Tanent

TANENT MANAGEMENT

Tanent NID

Tanent Name

City

Sector

Flat_ID

ADD UPDATE DELETE

Information Of Tanents

	t_id	t_name	city	sector	f_id
▶	2344623	moin	dhaka	12	1B
	T02	jerin	frangate	8	1A
	t05	jerin	frangate	8	1B
*					

15. Add Worker



Safe Home Authority

Worker MANAGEMENT

Worker NID

233

WorkerName

Moti

City

Dhaka

Sector

03

Mobile No

2343455

Flat_ID

1A

ADD

UPDATE

DELETE

Information Of Workers

	Worker ID	Worker Name	Worker City	Worker Sector	Worker Mobile No	Flat ID
▶	W01	ananta	tongi	0	98761	1A

16. Owner Show Information



Safe Home Authority

Search :

Search

	OWNER ID	Building ID	Building Name	Owner City	Owner Sector	Flat ID
▶	2020	07	AIUB	Dhaka	02	1A
	2020	07	AIUB	Dhaka	02	1A
	2020	07	AIUB	Dhaka	02	1B
	2020	07	AIUB	Dhaka	02	1B
*						

17. Developers



18. Logout From Owner Dashboard

OWNER DASHBOARD

Manage Your Building

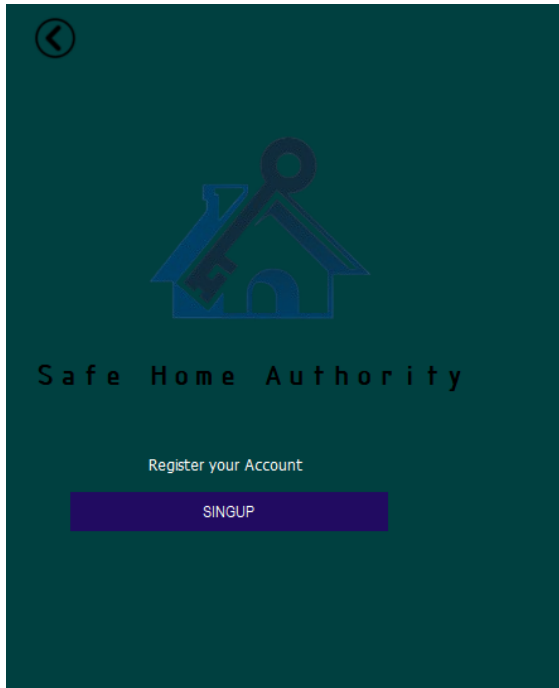
Building Id:

Success
Logout successful!
OK

List

	Owner ID	Name	Building ID	Building Name	Building Address City	Building Address Sector
▶	2020	Tamim	07	AIUB	Dhaka	02
*						

19. Police Login Form



X

Police Login Account

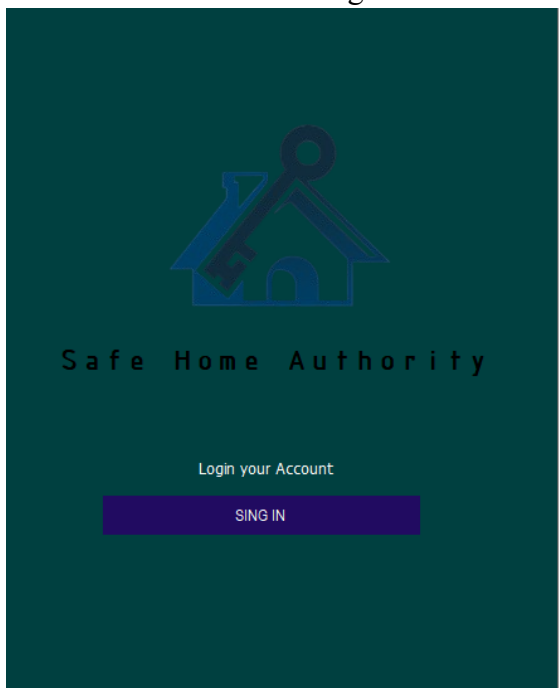
Police Id:

Password:

☐ Show Password

LOGIN

20. Police Registration Form



X

Register Account

User Id:

Password:

☐ Show Password

Job Title:

IGP
IGP
DIG
SP
ASP
Inspector
SI
ASI
Constable

21. Police Dashboard



Safe Home Authority

Police Information

?

?

MANAGEMENTINFORMATION

	Owner_ID	Owner_name	Mobile_number	Building_ID	Building_name	Building_city
▶	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2021	Anan	88016587127	08	NSU	Ghatpar
	2021	Anan	88016587127	08	NSU	Ghatpar
	2023	Anan	9178654	100	Ahsan Manjil	Dhaka
*						

22. Police Information



Safe Home Authority

Police Information

	p_id	p_name	jobtitle
▶	56	1234	DIG
*			

23. For Upcoming New Feature

policedashboard

POLICE DASHBOARD

Safe Home Authority

Police Information

?

?

MANAGEMENTINFORMATION

	Owner_ID	Owner_name	Mobile_number	Building_ID	Building_name	Building_city
▶	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2021	Anan	88016587127	08	NSU	Ghatpar
			88016587127	08	NSU	Ghatpar
*			9178654	100	Ahsan Manjil	Dhaka

Feature will available soon

OK

24. Logout From Police Dashboard

policedashboard

POLICE DASHBOARD

Safe Home Authority

Police Information

?

?

MANAGEMENTINFORMATION

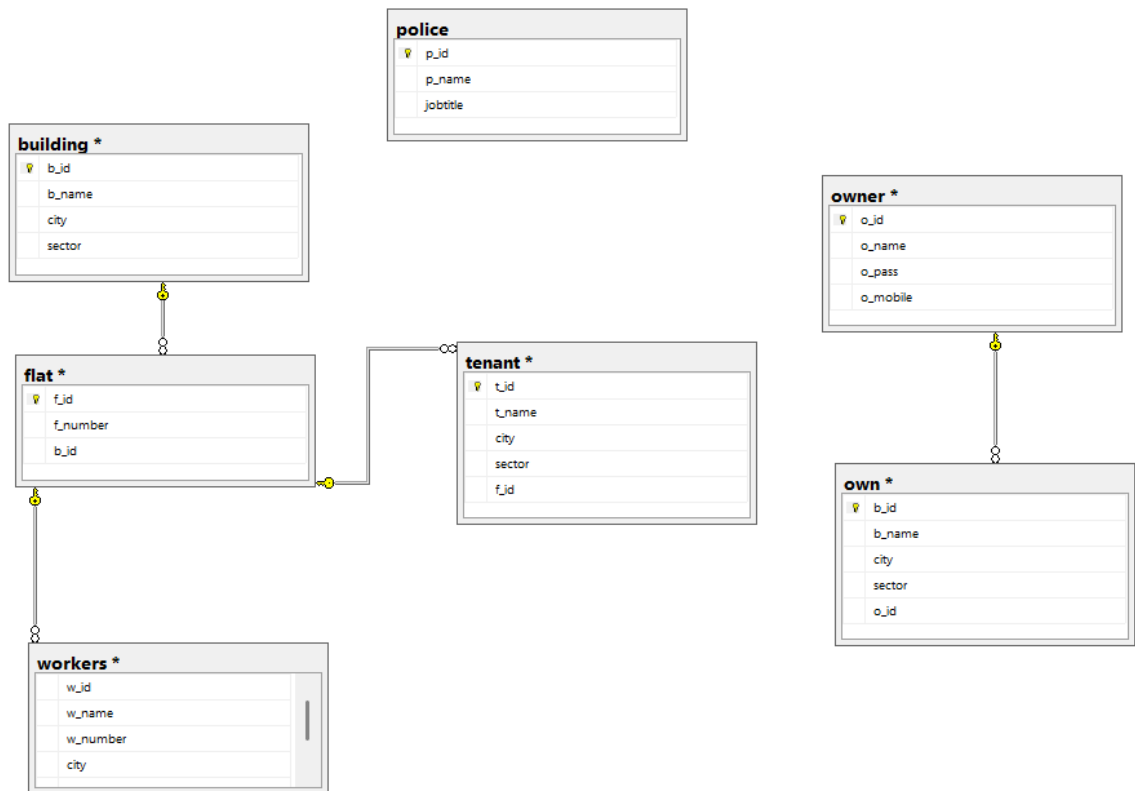
	Owner_ID	Owner_name	Mobile_number	Building_ID	Building_name	Building_city
▶	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2020	Tamim	8801709122214	07	AIUB	Dhaka
	2021	Anan	88016587127	08	NSU	Ghatpar
			88016587127	08	NSU	Ghatpar
*			9178654	100	Ahsan Manjil	Dhaka

Success

Logout successfull!

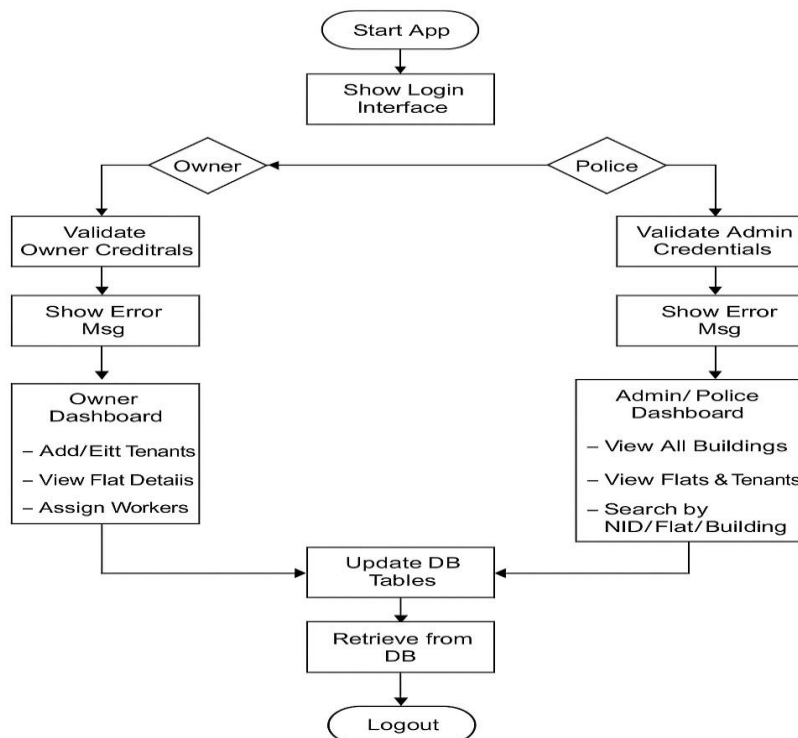
OK

Schema diagram:



Activity Diagram

Activity Diagram: Safe Home Authority System



Conclusion:

The Safe Home Authority system offers a secure and efficient platform to manage tenant information by bridging the communication gap between building owners and law enforcement. By implementing role-based access controls, the system ensures that building owners can maintain accurate tenant records for their properties, while police officers can access verified data across all registered buildings to support safety and investigation efforts. This centralized, digitally managed solution enhances transparency, accountability, and community security, making residential areas safer and more organized. Overall, the project demonstrates how technology can facilitate better coordination and trust between citizens and authorities in managing housing and tenant data.